

Presentation Script for Project presentation competition (4 presenters)

Khushi

Slide : (Title + Introduction to OBE)

Good morning....respected panel members and everyone present here. We are from Sandip polytechnic Department of Computer engineering.

Today here to present our project titled “**OBE Tracking System**”, Which is an automated solution to track outcomes with ease

Now the question arises in everyones mind is what is OBE tracking? So...

Outcome-Based Education, or OBE, is a structured educational framework that focuses on defining and achieving measurable learning outcomes.

Unlike traditional education systems, which mainly focus on syllabus coverage and exam completion, OBE emphasizes what the student is able to understand, apply, and demonstrate after completing a course or an academic program.

In OBE, learning outcomes are categorized into three levels:

- **Course Outcomes (COs)**, which specify the knowledge or skills a student should gain from a particular course which is basically subjects learned in each semester
- **Program Outcomes (POs)**, which represent broader abilities such as problem-solving, teamwork, and ethics which gain at the end of program
- **Program Specific Outcomes (PSOs)**, which focus on domain-specific technical skills

These outcomes are not just defined but must also be **mapped, assessed, and continuously evaluated** to ensure academic quality.

Since OBE focuses on measurable results, it requires a systematic approach to data collection and analysis.

This brings us to the role of accreditation bodies, which my teammate will now explain.

Gayatri

Slide : (Accreditation Requirements + Attainment Overview)

Accreditation bodies such as **NBA and NAAC** require institutions to strictly follow the OBE framework.

As part of accreditation, institutions must maintain:

- Proper CO–PO mapping
- Transparent and systematic attainment calculations

- Continuous Quality Improvement documentation
- Well-organized, audit-ready academic records

To measure these outcomes, OBE uses **attainment calculations**, which quantify how well students are achieving defined outcomes.

now lets understand about attainment ---

Attainment is classified into two levels:

Course level & program level

=====Co attainment

- In which course level had 2 assessments 1.Direct & Indirect
tools for direct assessment ...{ }
- **Direct Attainment**, which measures students' academic performance through exams and assessments
- where **Indirect Attainment**, which measures perceptions and feedback from students
- Indirect assessment tools include....{ CES which is conducted after the completion of each course }
to calculate Co direct attainment ..where in direct is divided into External & internal considered 50 % of internal & 50% of external

The formula is...it is caculated for each direct assessment tool ..the average of each tool is added as a Direct co and in the

same way for indirect the sum of survey rating score of CES is divided by the maximum possible score ...multiplied by 100

And taking 80% of Direct and 20% indirect Co the final CO attainment is calculated...

=====Po attainment

for program level attainment again dived into Direct and Indirect..where Direct PO is derived from overall CO attainment of mapped courses and indirect PO attainment tools are..{}

Formula explanation----

Direct Po Attainment taken by ...Sum of final Co attainment and Co-po weight which is divided by maximum weight

Where Indirect attainment score is taken by average rating of all indirect tools..

as we calculate final CO the final Po is also taken by 80% of direct Po and 20% of indirect Po

Similar to po attainment calculation we calculate PSO attainment by using PSo values

These two types of attainment together provide a **balanced evaluation**, combining actual performance with feedback-based evaluation.

The final attainment values help institutions analyze gaps and take corrective actions for improvement.

Now, my teammate will explain the detailed assessment tools used to calculate attainment.

Farheen

Slide : (Assessment Tools + CO & PO Attainment Process)

At the **course level**, direct assessment is based on students' academic performance using multiple tools to ensure fairness and coverage.

These tools include:

- **Formative Assessment Theory**, such as class tests, which help evaluate continuous learning
- **Formative Assessment Practical**, including lab manuals and experiments
- **Self-Learning Assessment**, such as assignments or micro-projects
- **Summative Assessment Theory**, which is the final theory examination

- **Summative Assessment Practical**, which evaluates external practical exam...

Indirect assessment at the course level is measured using the **Course Exit Survey**, where students rate predefined statements based on their learning experience.

Using both direct marks and indirect survey ratings, **Course Outcome attainment is calculated.**

At the **program level**:

- **Direct PO attainment** is derived from the overall CO attainment of mapped courses
- **Indirect PO attainment** is calculated using Program Exit Surveys, Alumni feedback, Resource Person feedback, and feedback from curricular and extra-curricular activities

These values are combined using weighted formulas to calculate final PO attainment.

Currently, managing all these calculations manually is challenging.

Saloni

Slide : *(Existing System + Limitations)*

In the existing system, most institutions manage OBE

activities using manual documentation and spreadsheet tools such as Microsoft Excel.

Faculty members are responsible for:

- **Creating CO–PO–PSO mappings**
- **Entering marks for different assessments**
- **Manually calculating attainment values**
- **Preparing reports for accreditation**

While this method is commonly used, it becomes difficult to manage as the number of students, courses, and outcomes increases.

The system faces several challenges:

- **Data inconsistency due to different spreadsheet formats**
- **Higher chances of manual calculation errors**
- **Difficulty in preparing audit-ready reports within limited time**
- **Lack of proper data security and access control**
- **Increased workload on faculty members**

These challenges highlight the need for an automated and centralized solution.

To address this, we propose an OBE Tracking System, which my teammate will now explain.

Khushi (Again)

Slide:(Proposed System + Data Flow Diagram)

Our proposed solution is an OBE Tracking System, which is an automated web-based platform designed to manage all Outcome-Based Education activities efficiently.

The system automates:

- **CO–PO–PSO mapping // not automated**
- **Attainment calculations**
- **Report generation**

Faculty can directly enter CIS and assessment marks into the system.

An important feature of our system is that indirect survey responses are digitally collected, stored, and processed automatically, eliminating the need for faculty to manually enter survey scores again.

The system maintains a centralized academic database, ensuring consistency and easy access.

The data flow diagram illustrates how data flows from users to the system, is processed by the backend, stored securely, and used to generate attainment results and reports.

Now, my teammate will explain the system architecture and technologies used.

Farheen and Gayatri(Again)

Slide:(System Architecture + Technology Stack)

The proposed system follows a client–server architecture.

The frontend provides a user-friendly interface for data entry, visualization, and report access.

The backend handles business logic, validation, security, audit logging, and data processing.

A centralized database stores course details, outcome mappings, assessment data, attainment results, user information, and generated reports.

This architecture ensures scalability, data consistency, accountability, and ease of maintenance.

Where the ...

The technology stack used includes:

- React and Bootstrap for frontend development
- Django REST Framework for backend services
- PostgreSQL and SQLite for database management

Next, my teammate will discuss the advantages and future scope of the system.

Saloni (Again)

Slide:(saloni-Advantages + Applications Gayatri-Future Scope + Khushi-Conclusion)

The proposed system offers several advantages over the existing approach.

It significantly reduces manual workload by automating attainment calculations.

It improves accuracy by minimizing human errors and ensures centralized data management.

Role-based access control enhances data security, while automated report generation supports audits and accreditation processes.

The system can be applied for standardized academic data management, performance monitoring, and continuous quality improvement.

In the future, the system can be enhanced by integrating:

- AI-based performance prediction and analysis
- Mobile application support
- Data-driven curriculum planning

To conclude, the OBE Tracking System simplifies and automates the Outcome-Based Education process, improves efficiency, and provides reliable, audit-ready academic data.

Thank you for your time and attention.

